The MiniBooNE/LSND Anomaly and Sterile Neutrino Exclusion from $D = 3$

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

March 2026

Abstract

We analyze the MiniBooNE/LSND anomaly within Recognition Science (RS). Both experiments report excess $\bar{\nu}_e$ appearance in ν_μ beams, consistent with neutrino oscillations at $\Delta m^2 \sim 1$ eV^2—a mass splitting incompatible with the standard three-neutrino framework. The conventional explanation invokes a sterile neutrino with $m_4 \sim 1$ eV. RS *excludes* this interpretation: the spatial dimension $D = 3$ forces exactly three neutrino generations via the face-pair count of the recognition cube, $N_\nu = \text{face_pairs}(3) = 3$, leaving no structural slot for a fourth mass eigenstate. We further propose that the φ-ladder spacing of the three active neutrino masses ($m_i \propto \varphi^{n_i}$) produces L/E-dependent interference modulations that can, at specific baselines, mimic a sterile oscillation signal. This mechanism is a hypothesis, not a proved theorem; it is offered as a falsifiable alternative to sterile neutrino models. The MicroBooNE experiment (2021) has demonstrated that a significant fraction of the MiniBooNE excess can be explained by standard-model backgrounds (radiative $\Delta \to N\gamma$ decays); the RS analysis is consistent with this finding. The structural theorem ($N_\nu = 3$) follows from the RS forcing chain; the φ-aliasing hypothesis awaits full quantitative treatment. The Short-Baseline Neutrino (SBN) program at Fermilab—SBND, MicroBooNE, and ICARUS—provides the decisive experimental test.

1 Recognition Science Framework

The structural exclusion of sterile neutrinos follows from the RS forcing chain, which originates in the cost axioms.

Definition 1.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 1.2 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Set $H(t) = J_{\log}(t) + 1$. Axiom A2 transforms into the d'Alembert equation $H(s + t) + H(s - t) = 2H(s)H(t)$. By the Aczél classification theorem [2], the unique continuous solution with $H(0) = 1$ and $H''(0) = 1$ is $H(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

The recognition cube Q_D has spatial dimension $D = 3$ forced by $2^D = 8$ (the 8-tick epoch) and three independent dimension-forcing arguments. Its face-pair count face_pairs(3) = 3 is the structural origin of exactly three active neutrino generations. The φ-ladder ($\varphi = (1 + \sqrt{5})/2$ forced by RCL self-similarity) assigns masses $m_i \propto \varphi^{n_i}$ to the three mass eigenstates. No fourth mass eigenstate is admitted by the RS cube structure.

2 Introduction

The three-neutrino oscillation framework successfully describes solar, atmospheric, and long-baseline reactor data with two independent mass splittings: $\Delta m^2_{21} \approx 7.5 \times 10^{-5}$ eV^2 (solar) and $|\Delta m^2_{32}| \approx 2.5 \times 10^{-3}$ eV^2 (atmospheric) [3].

Two experiments report anomalous $\bar{\nu}_e$ appearance that does not fit this framework:

- **LSND** (1996–2001): excess $\bar{\nu}_e$ in a $\bar{\nu}_\mu$ beam from μ^+ decay at rest; $L \approx 30$ m, $E \sim 20$–200 MeV; $\Delta m^2 \sim 1$ eV^2 [4].
- **MiniBooNE** (2018, 2020): excess low-energy ν_e-like events in a ν_μ beam at Fermilab; $L \approx 500$ m, $E \sim 200$–3000 MeV; combined significance $\sim 4.8\sigma$ after 12.84×10^{20} protons on target [5].

The combined LSND+MiniBooNE significance exceeds 6σ. The canonical new-physics interpretation is a sterile neutrino with $m_4 \sim 1$ eV and mixing $|U_{e4}|^2 \sim 10^{-3}$, forming a "3+1" oscillation model. This is in tension with:

- Muon-neutrino disappearance bounds (IceCube, MINOS+).
- Cosmological limits on N_{eff} and the sum of neutrino masses.
- MicroBooNE's identification of radiative backgrounds in the same energy region [6].

The status of the anomaly is therefore *ambiguous*: it may reflect an uncontrolled background, a systematic error, or genuine new physics incompatible with the standard three-neutrino picture. The RS analysis [1] below offers a structural argument against sterile neutrinos and a qualitative model for residual excess events.

3 The Structural Argument: $N_\nu = 3$ from $D = 3$

Theorem 3.1 (Exactly Three Neutrino Generations)**.** *In Recognition Science, the spatial dimension $D = 3$ forces exactly three active neutrino generations:*

$$N_\nu = \text{face_pairs}(D) = D = 3. \tag{1}$$

A fourth (sterile) neutrino is excluded by the spinor structure of the D-dimensional recognition cube.

Proof. The recognition cube Q_D has $2D$ faces organized into D orthogonal face pairs (top/bottom, front/back, left/right for $D = 3$). Each face pair generates one independent chiral spinor mode; these are the three active neutrino flavors ν_e, ν_μ, ν_τ. With $D = 3$:

face_pairs(3) = 3. There is no fourth face pair; hence there is no structural slot for a fourth light neutrino species. □

Corollary 3.2. face_pairs(3) $\neq 4$*: a sterile neutrino at* $m_4 \sim 1$ *eV is excluded by RS. The correct prediction from RS is* $N_\nu = 3$ *active flavors with masses on the* φ*-ladder.*

Remark 3.3. The dimension $D = 3$ is itself forced in RS by three independent arguments derived from the RS forcing chain [1]: (i) Alexander duality requires integer linking invariants only in $D = 3$; (ii) Kepler stability of near-circular orbits requires $D = 3$; (iii) the minimum dyadic-to-sexagesimal synchronization (8-tick and 45-tick cycles) achieves $\mathrm{lcm}(8, 45) = 360$ exactly at $D = 3$.

4 The φ-Aliasing Hypothesis

The exclusion of sterile neutrinos raises the question: what produces the LSND/MiniBooNE excess? We propose, as a hypothesis to be tested quantitatively, the following mechanism.

Proposition 4.1 (φ-Ladder Neutrino Masses)**.** *In RS, the three active neutrino masses sit on the* φ*-ladder:*

$$m_i \approx m_0\, \varphi^{n_i}, \quad n_1 < n_2 < n_3, \tag{2}$$

where m_0 *is a reference mass set by the coherence energy scale and* $\varphi = (1 + \sqrt{5})/2$.

Definition 4.2 (φ-Aliasing)**.** At baseline L and energy E, the two-flavor oscillation probability $P(\nu_\mu \to \nu_e)$ receives contributions from all three mass splittings Δm^2_{ij}. For φ-spaced masses, higher rungs give $\Delta m^2_{3i} \sim \varphi^{2(n_3-n_i)} \Delta m^2_{21}$. At baselines L/E tuned to a high rung splitting, the summed oscillation probability can exhibit a beat frequency that is numerically consistent with a *new* splitting $\Delta m^2_{\mathrm{eff}} \sim 1\ \mathrm{eV}^2$, without introducing a fourth eigenstate.

Remark 4.3. This mechanism is qualitatively analogous to aliasing in signal processing: the discrete φ-ladder spacing can create apparent "extra" oscillation frequencies in a finite baseline window. However, the analogy requires a quantitative calculation to determine whether the φ-spaced active masses can actually produce an apparent $\Delta m^2 \sim 1\ \mathrm{eV}^2$ at the LSND and MiniBooNE baselines. This calculation is *not* completed in the present paper and represents an open question. The hypothesis is stated here as a target for future work.

5 MicroBooNE and Background Explanations

The MicroBooNE experiment [6] operated in the same neutrino beam as MiniBooNE with a liquid-argon time-projection chamber that allows event-by-event particle identification. Key findings:

- MicroBooNE found no excess of charged-current quasi-elastic ν_e events consistent with a simple sterile neutrino oscillation.
- Radiative $\Delta^+ \to p + \gamma$ decays, in which the photon mimics an electron in the MiniBooNE Cherenkov detector, account for a substantial fraction of the observed excess in the MiniBooNE signal region.

- A residual excess may remain in some MicroBooNE channels, but its nature and significance continue to be analyzed.

Theorem 5.1 (RS Consistency with Background Explanations)**.** *If the MiniBooNE excess is entirely due to mis-identified $\Delta \to N\gamma$ and π^0 backgrounds, the three-generation RS prediction is confirmed: no fourth neutrino eigenstate exists.*

Proof. The background explanation requires only standard-model processes and the three active neutrinos. This is entirely consistent with $N_\nu = 3$ (Theorem 3.1). □

6 Comparison with Other Explanations

Explanation	**Prediction**	**Status**
Sterile neutrino (3+1)	ν_4 with $m_4 \sim 1$ eV	Tension with IceCube, MINOS+, N_{eff}; excluded by RS ($N_\nu = 3$)
SM backgrounds (Δ, π^0)	No fourth mass eigenstate	Supported by MicroBooNE; consistent with RS
Lorentz violation	CPT-odd oscillations	No evidence; not predicted by RS
RS φ-aliasing	Three-active-ν beat frequency	Hypothesis; quantitative calculation needed

7 Predictions and Falsifiers

1. **No sterile neutrino discovery.** The SBN program (SBND, MicroBooNE, ICARUS) will not find evidence for a fourth neutrino mass eigenstate. Any residual excess will be consistent with backgrounds or the φ-aliasing mechanism, not a new oscillation frequency. *Falsifier*: a confirmed oscillation minimum at the predicted Δm^2_{14} in a disappearance channel.

2. $N_{\text{eff}} = 3.044$**.** The effective number of thermalized neutrino species remains at the standard three-generation value. A measurement of $N_{\text{eff}} > 3.1$ at $> 5\sigma$ would challenge RS.

3. **φ-aliasing test.** If the φ-aliasing hypothesis is developed quantitatively, it predicts a specific L/E dependence for any residual excess—one consistent with an interference pattern of three φ-spaced oscillation frequencies, not a simple two-frequency pattern. High-statistics SBN data can distinguish these.

4. **Background-only explanation.** If MicroBooNE's full analysis and the SBND data find that all excess events are accounted for by known backgrounds, the RS three-generation prediction stands.

8 Conclusion

The MiniBooNE/LSND anomaly remains *ambiguous* pending final SBN results. The RS analysis provides one firm structural conclusion: there is no fourth neutrino mass eigenstate. The dimension $D = 3$ of the recognition framework forces exactly three active neutrino generations, and a sterile neutrino at $m_4 \sim 1$ eV is structurally excluded. The φ-aliasing hypothesis offers a qualitative mechanism for residual excess events, but quantitative derivation remains as future work. MicroBooNE data supporting background explanations are consistent with the RS prediction. The SBN program will provide the definitive test.

References

[1] J. Washburn, "The Algebra of Reality: A Recognition Science Derivation of Physical Law," *Axioms* **15**(2), 90 (2026). doi:10.3390/axioms15020090

[2] J. Aczél, *Lectures on Functional Equations and Their Applications* (Academic Press, 1966).

[3] S. Navas *et al.* (Particle Data Group), "Review of Particle Physics," Phys. Rev. D **110**, 030001 (2024).

[4] A. Aguilar et al. (LSND Collaboration), "Evidence for neutrino oscillations from the observation of $\bar{\nu}_e$ appearance in a $\bar{\nu}_\mu$ beam," Phys. Rev. D **64**, 112007 (2001).

[5] A. A. Aguilar-Arevalo et al. (MiniBooNE Collaboration), "Updated MiniBooNE neutrino oscillation results with increased data and new background studies," Phys. Rev. D **103**, 052002 (2021).

[6] P. Abratenko et al. (MicroBooNE Collaboration), "Search for an anomalous excess of inclusive charged-current ν_e interactions in the MicroBooNE experiment using Wire-Cell reconstruction," Phys. Rev. D **105**, 112005 (2022).